

Silver Ions (Ag⁺) Action on Microbial Cells

Source: Chaloupka et al., *Trends in Biotechnology*, 2010

1. Introduction

Silver ions (Ag⁺) = powerful antimicrobial agents. Act on bacteria via **multiple simultaneous mechanisms** → difficult for bacteria to develop resistance. Targets: cell wall, membrane, ribosomes, DNA.

2. Mechanisms of Action (4 Key Targets)

A. Destruction of Cell Wall & Cell Membrane Lysis

- Ag⁺ attack **peptidoglycan layer** of bacterial cell wall
- Peptidoglycan = structural backbone of gram-positive & gram-negative bacteria
- Ag⁺ disrupt cross-linking → wall weakens → **cell lysis** (rupture)
- Membrane integrity lost → cell contents leak → cell death

B. Ribosome Denaturation → Inhibition of Protein Synthesis

- Ag⁺ bind to **70S ribosomes** (prokaryotic ribosome type — unique to bacteria, NOT found in human cells)
- Denaturation = ribosome loses 3D shape → **non-functional**
- No functional ribosome → **no protein synthesis**
- Without proteins: enzymes fail, membrane degrades further, cell cannot survive
- Also causes **plasma membrane degradation** as a downstream effect

C. DNA Condensation → Block Replication

- Ag⁺ bind directly to **DNA bases** (nitrogenous bases: A, T, G, C)
- Binding → DNA **condenses** (compacts abnormally)
- Condensed DNA **cannot be read or copied** by DNA polymerase
- Result: **Binary fission blocked** — bacteria cannot reproduce
- Also affects **DNA plasmids** (extra-chromosomal DNA rings shown in diagram)

D. Penetration via Ion Diffusion

- Ag⁺ ions are small, charged → penetrate through damaged/disrupted membrane
- Once inside → attack all 3 internal targets simultaneously (ribosomes, DNA, plasmid)

3. Structures Targeted (from diagram)

Structure	Role in Bacteria	Ag ⁺ Effect
Bacterial cell wall	Structural support	Destroyed → lysis
Cell membrane	Selective barrier	Degraded → leakage
70S ribosome	Protein synthesis	Denatured → no proteins
DNA plasmid	Extra genes, resistance	Condensed → non-functional
Chromosomal DNA	Replication & function	Condensed → no binary fission

4. Why Ag⁺ is Effective (Multi-Target Strategy)

Multi-target attack = key advantage:

- Single drug → single target → bacteria mutate one gene → resistance
 - Ag⁺ → attacks **4 targets at once** → bacteria would need simultaneous mutations in all 4 pathways → extremely unlikely → **low resistance development**
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5. Selectivity (Why Safe for Humans)

- Human cells = **eukaryotic** → have **80S ribosomes** (not 70S) → Ag⁺ ribosome effect less applicable
 - Human cells lack peptidoglycan cell wall → cell wall attack irrelevant
 - Still some mammalian toxicity at high doses → used topically/externally (wound dressings, medical devices)
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6. Applications

- **Wound dressings** (burns, chronic wounds)
 - **Medical device coatings** (catheters, implants) → prevent biofilm
 - **Water purification**
 - **Nanoparticle delivery** (silver nanoparticles = sustained Ag⁺ release)
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7. Summary (10-mark answer structure)

Silver ions exert antimicrobial action via 4 simultaneous mechanisms: **(1)** Destroy peptidoglycan cell wall → membrane lysis **(2)** Denature 70S ribosomes → inhibit protein synthesis → membrane degradation **(3)** Bind DNA bases → condense DNA → block replication/binary fission **(4)** Damage plasmid DNA → loss of accessory gene function Multi-target nature = high efficacy + low resistance risk. Selectivity for prokaryotic structures = therapeutic advantage.

Keyword list for exam: Ag⁺, peptidoglycan, lysis, 70S ribosome, denaturation, protein synthesis, DNA condensation, binary fission, plasmid, multi-target antimicrobial.